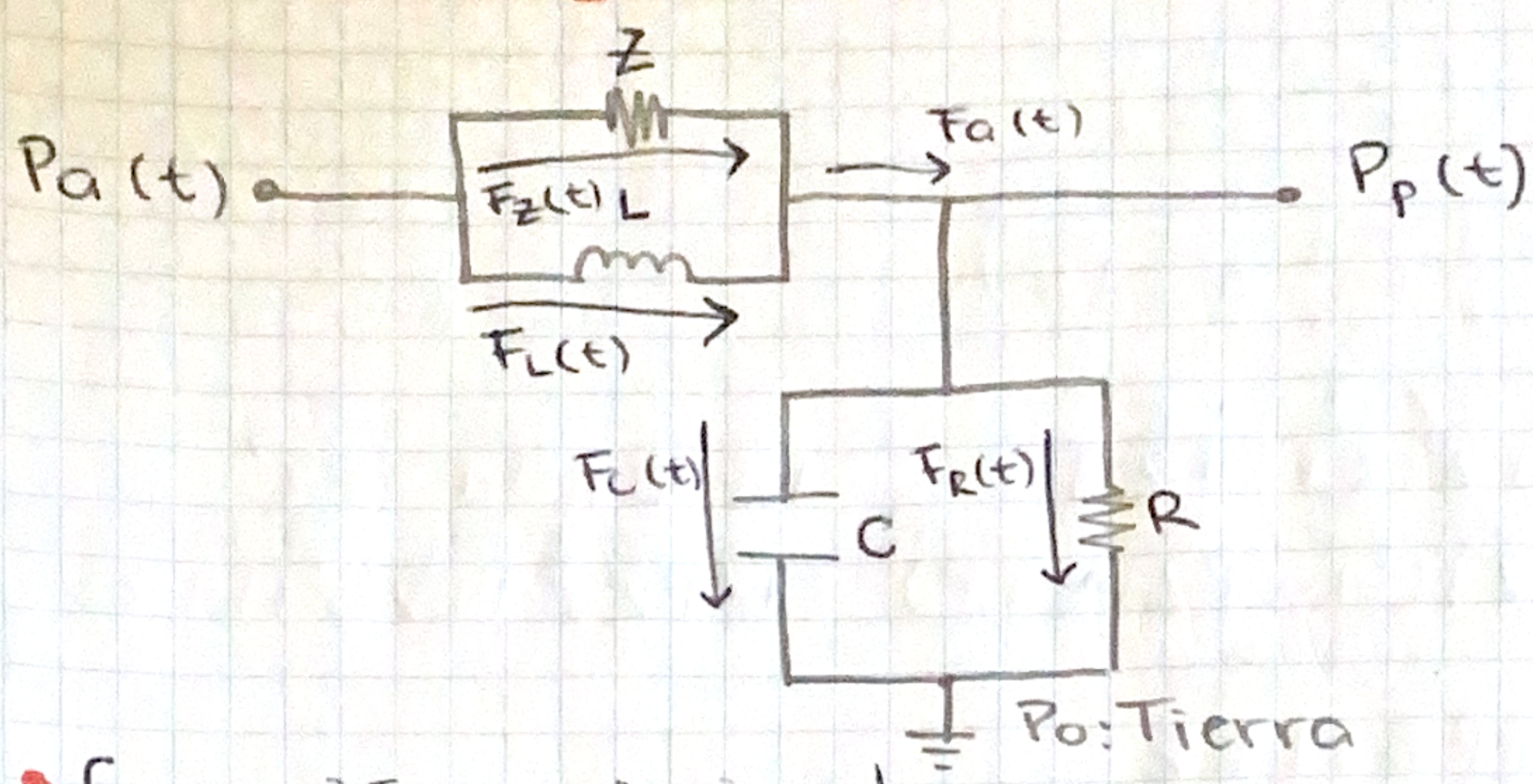
w

Práctica 5.4-Sistema Cardiovascular

10-10-25



- Ecuación principal

$$F_a(t) = F_Z(t) + F_L(t) = F_C(t) + F_R(t)$$

$$F_Z(t) = \frac{P_a(t) - P_p(t)}{Z}$$

$$F_L(t) = \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt$$

$$F_C(t) = \frac{C\,dP_p(t)}{dt} \qquad F_R(t) = \frac{P_p(t)}{R}$$

- Sustituir en la ecuación principal para Procedimiento algebraico

$$\frac{P_a(t)}{Z} - \frac{P_p(t)}{Z} + \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt = \frac{C\,dP_p(t)}{dt} + \frac{P_p(t)}{R}$$

Transformada de Laplace

$$\frac{P_a(s)}{Z} - \frac{P_p(s)}{Z} + \frac{P_a(s) - P_p(s)}{LS} = CS\,P_p(s) + \frac{P_p(s)}{R}$$

$$\left(\frac{1}{Z} + \frac{1}{LS}\right)P_a(s) = \left(CS + \frac{1}{R} + \frac{1}{Z} + \frac{1}{LS}\right)P_p(s)$$

$$\frac{CS}{1}+\frac{1}{R}=\frac{CSR+1}{R}+\frac{1}{Z}=\frac{CSRZ+Z+R}{RZ}+\frac{1}{LS}=\frac{CLS^2RZ+Z+R}{LSRZ}$$

• Función de transferencia $\frac{P_p(t)}{P_a(t)}$

$$\frac{P_p(t)}{P_a(t)}=\frac{\left(\frac{1}{Z}+\frac{1}{LS}\right)}{\left(CS+\frac{1}{R}+\frac{1}{Z}+\frac{1}{LS}\right)}=\frac{\frac{LS+Z}{LSZ}}{\frac{CLS^2RZ+Z+R}{LSRZ}}$$

$$=\frac{(LS+Z)(LSRZ)}{(LSZ)(CLS^2RZ+Z+R)}=\frac{LS^2RZ+LSRZ^2}{CL^2RZ^2S^3+LZ^2S+LRZS}$$

$$=\frac{LZS(RS+RZ}{LZS(CLRZS^2+}$$

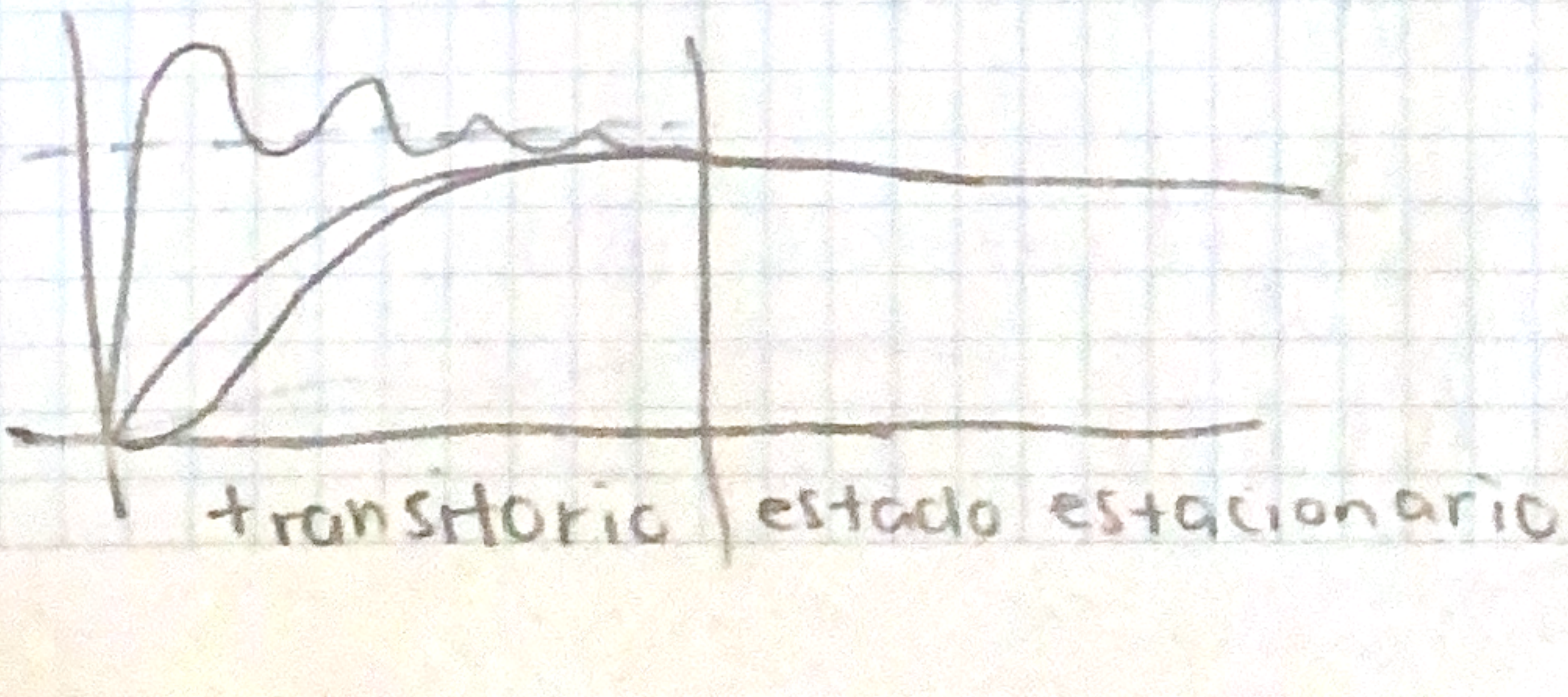
$$\frac{P_p(s)}{P_a(s)}=\frac{\frac{LS+Z}{LZS}}{\frac{CLS^2+(LZ+RL)S+RZ}{RLZS}}=\boxed{\frac{RLS+RZ}{CLRZS^2+(LZ+RL)S+RZ}}$$

• Error en estado estacionario

$$e(s)=\lim_{s\to 0} s\,P_a(s)\left[1-\frac{P_p(s)}{P_a(s)}\right]$$

$$=\lim_{s\to 0}\cancel{s}\,\frac{1}{\cancel{s}}\left[1-\frac{RLS+RZ}{CLRZS^2+(LZ+RL)S+RZ}\right]$$

$$=1-\frac{RZ}{RZ}=0V$$



- Estabilidad en lazo abierto

$$\lambda_{1,2} = -\frac{b \pm \sqrt{b^2-4ac}}{2a}$$

$a = CLRZ$

$b = LZ + RL$

$c = RZ$

$$\lambda_{1,2} = \frac{-(LZ+RL) \pm \sqrt{(LZ+RL)^2 - 4CLR^2Z^2}}{2CLRZ}$$

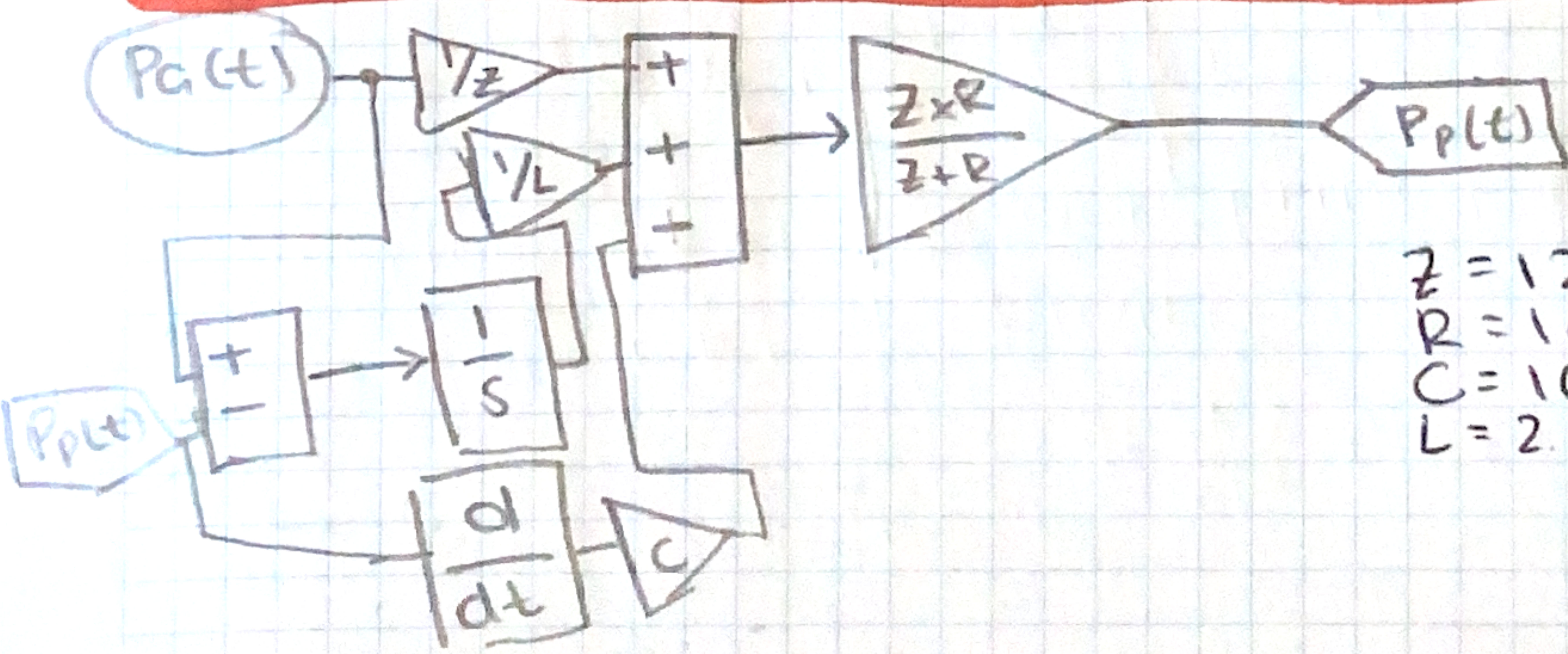
$$= \frac{(-)(\;)}{+}$$

→ El sistema tiene una respuesta estable porque $Re\,\lambda_{1,2} < \emptyset$

- Modelo de ecuaciones integrodiferenciales

$$P_p(t)\left(\frac{1}{R}+\frac{1}{Z}\right) = \left(\frac{P_a(t)}{Z}\right) + \frac{1}{L}\int\left[P_a(t)-P_p(t)\right]dt - \frac{C\,dP_p(t)}{dt}$$

$$\boxed{P_p(t) = \left(\frac{P_a(t)}{Z} + \frac{1}{L}\int\left[P_a(t)-P_p(t)\right]dt - \frac{C\,dP_p(t)}{dt}\right)\frac{ZR}{Z+R}}$$



$Z = 120\,k\Omega$
$R = 100\,k\Omega$
$C = 100\,\mu F$
$L = 2.2\,mH$